

Advait Deshmukh

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Boulder, CO, USA

RESEARCH INTERESTS

Natural Language Processing, Narrative Understanding, Computational Discourse, Interpretability

EDUCATION

Doctor of Philosophy, Computer Science

University of Colorado Boulder

Advised by Dr. Maria Antoniak

Jan 2026 – Dec 2030

Boulder, CO

Master of Science, Computer Science

University of Colorado Boulder

Advised by Dr. Maria Leonor Pacheco

Aug 2023 – Dec 2025

Boulder, CO

Bachelor of Engineering, Computer Science

Birla Institute of Technology and Science, Pilani

Aug 2017 – Aug 2021

Pilani, India

PUBLICATIONS

C=CONFERENCE, S=IN SUBMISSION

- [C.1] Advait Deshmukh, Nora Benedict, Melanie Walsh, and Maria Antoniak. 2026. The Garden of Forking Prompts: How Users Explore Narrative Space in Story Generation. In Proceedings of the Conference on Language Modeling (COLM), Oct 2026, San Francisco, CA, <https://arxiv.org/abs/2609.14677>.
- [C.2] Rohan Das, Advait Deshmukh, Alexandria Leto, Zohar Naaman, I-Ta Lee, and Maria Leonor Pacheco. 2026. A Structured Clustering Approach for Inducing Media Narratives. Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026), San Diego, CA, <https://aclanthology.org/2026.acl-long.1970/>.
- [C.3] Advait Deshmukh, Ashwin Umadi, Dananjay Srinivas, and Maria Leonor Pacheco. 2025. All Entities are Not Created Equal: Examining the Long Tail for Ultra-Fine Entity Typing. Proceedings of the 14th Joint Conference on Lexical and Computational Semantics (*SEM 2025), co-located with EMNLP 2025, <https://aclanthology.org/2025.starsem-1.15>.
- [S.1] Mreedul Gupta, Advait Deshmukh, Ashwin Umadi, Matt Pauk, and Maria Leonor Pacheco. 2026. Narrative-UFET: Narrative Generation for Ultra-Fine Entity Typing. arXiv preprint, [arXiv:2606.27598](https://arxiv.org/abs/2606.27598).

NON-ARCHIVAL PRESENTATIONS

- [N.1] Advait Deshmukh, Nora Benedict, Melanie Walsh, and Maria Antoniak. 2026. The Garden of Forking Prompts: How Users Explore Narrative Space in Story Generation. TADA 2026, Berkeley, CA.

SELECTED RESEARCH PROJECTS

How Users Explore Narrative Space in Story Generation

Advised by Dr. Maria Antoniak

Spring 2026 – Summer 2026

CLS Lab, CU Boulder

- Constructed WildStories, a sample of 275,635 story generation prompts, and WildEdits, 24,291 edit trees tracing how users iteratively revise prompts, from naturally occurring user-chatbot conversations in WildChat.
- Developed a framework of edit types crossing four directions with fourteen narrative targets, and showed that automated permutations of it can be used to benchmark story generation.

A Structured Clustering Approach for Inducing Media Narratives

Collaborator

Spring 2025 – Summer 2025

BLAST Lab, CU Boulder

- Implemented a modified DAPrompt (Xiang et al.) based relationship classifier to extract causally related event pairs across 4 datasets.
- Infused knowledge from teacher models (Llama3, Qwen3) to aid in domain adaptation.

Examining the Long Tail for Ultra-Fine Entity Typing

Independent study advised by Dr. Maria Pacheco

Fall 2024 – Spring 2025

BLAST Lab, CU Boulder

- Conducted rigorous empirical analysis to study limitations of predominantly popular language model-based approaches for ultra-fine entity typing.
- Developed proxies to predict entity frequencies based on pretrained language models and analyzed its effect on typing performance.

TEACHING EXPERIENCE

University of Colorado Boulder

Teaching Assistant, CSCI 1300: Intro to Computing

Fall 2024 – Spring 2025

Boulder, CO

- Assisted in the course by grading, holding recitations & office hours to support students.
- Collaborated with faculty to redesign course materials, improving student engagement.

University of Colorado Boulder

Course Manager, CSCI 1300: Intro to Computing

Fall 2023 – Spring 2024

Boulder, CO

- Managed operations for the course, serving over 900 students through office hours and online platform Ed.
- Collaborated with faculty to create and grade assignments, projects & exams.

PROFESSIONAL EXPERIENCE

Amazon

Applied Science Intern

Summer 2026

Bellevue, WA

- Developed an atomic claim evaluation (ACE) system to evaluate agentic systems for the Supply Chain Optimization Technologies (SCOT) long-term planning & forecasting team.
- Improved error detection precision by 30% and recall by 50% over an LLM-as-judge baseline, across both offline evaluation against human outputs and live evaluation against source documents.

Societe Generale Global Solution Centre

Software Development Engineer

Jul 2021 – Jul 2023

Bangalore, India

- Developed a Retrieval Augmented Generation (RAG) based chat system to assist support teams in efficiently handling policy queries.
- Built an innovation portal, generating 120 ideas across 7 departments to foster cross-departmental collaboration.

Centre for Development of Advanced Computing (CDAC)

Research Intern

Jul 2020 – Dec 2020

Pune, India

- Engineered a scalable platform for High Performance Computing (HPC) job submission to optimize data processing.
- Developed a portal for seamless management of big data tasks, enhancing research efficiency.

VoiceQube

Summer Intern

May 2019 – Jul 2019

Bangalore, India

- Participated in summer internship program as part of the practice school program at BITS Pilani.
- Developed a conversational service to assist students in their GRE preparation.

SKILLS

Programming Languages: Python, JavaScript, TypeScript, C++, Java

Frameworks & Tools: React, Spring, FastAPI, Flask

Machine Learning Libraries: PyTorch, Transformers, NLTK, Pandas, Numpy, Langchain, Ollama

Cloud Technologies: Google Cloud, AWS, Microsoft Azure

DevOps & Version Control: Docker, Git, Kubernetes

AWARDS AND GRANTS

Conference Support Funds – \$1500

Computer Science Department, University of Colorado

Dec 2025

Boulder, USA

Travel Subsidy Recipient – \$750

*SEM 2025, co-located with EMNLP 2025

Nov 2025

Suzhou, China

Lloyd Botway Fellow – \$500

Computer Science Department, University of Colorado

April 2025

Boulder, USA

Outstanding Teaching Assistant – \$500

CSCI 1300, University of Colorado

March 2025

Boulder, USA

Winner - Poster Competition – \$500

Annual Research Expo, University of Colorado

Feb 2025

Boulder, USA

Grad App Aid Recipient – \$500

QueerInAI

Nov 2024

Boulder, USA

LEADERSHIP EXPERIENCE

Founder and President

VoiceQube Pilani

Sep 2019 – May 2021

- Founded and led VoiceQube BITS Pilani Chapter to develop Voice interface-based projects.
- Deployed BITS Buddy to handle queries about campus life, responding to ~4000 requests between January and March 2020.